SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)

DepartmentofComputerScienceandEngineering

ASSESSMENTTOOL2–TraversalChallenge

CourseCode:	CSA0305	**CourseTitle:**	DataStructures
COAssessed:	CO3–Precision(BL3,BL5)	**Assessment:**	AssessmentTool2–Traversal Challenge
Weightage:	30%ofCIA	**TotalMarks:**	100
CO3Statement:	*SolveproblemsforoptimizeddatastorageandretrievalusingBinarySearchTrees,AVLTrees,B-Trees, Red-Black Trees, and Splay Trees.*		
Student Name:	H SARAVANA	**Reg. No.:**	192511397
Section/Batch:	CSE	**Date:**	03/08/2026

CodeofConduct

I H SARAVANA (192511397)(Name/RegNo) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature ofthe Student: H saravana

Instructions

- Thisassessmenttoolcontains5TraversalChallengequestions,Total:100 Marks.
- Studentscanuseanyonline/offlineTraversalChallenge.Demonstratetree.
- Whenastudentanswerusingsimulator,inevaluationtheanswershouldhavecapturedscreenshots after every major operation
- SubmitthereportinPDFformatwithproperlylabelledscreenshotsasproofof completion.

Section/Topic	Focus	QNos	Marks
Tree Traversals	Traversaltypes,Inorder,PostorderandPreorder	5	100
	GRANDTOTAL:	**100Marks**	

TraversalChallenge

RubricsforCalculation

Criteria	Weightage	Level4–Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
ConceptualUnderstanding	20	Demonstratesthoroughunderstanding of all coreData Structures concepts;answersaccuratelyreflectdeepknowledge	Showsgoodunderstandingwith minorconceptualgaps	Basicunderstanding; severalmisconceptionspresent	Limitedunderstanding;majorconceptsmisunderstood
TraversalAccuracy	20	Correctlycompletesalltreetraversals(Preorder,Inorder, Postorder, LevelOrder) without errors.	Completesmosttraversalscorrectlywithoneminorerror.	Correctlyperformsonlysometraversalswithmultipleerrors.	Unabletoperformmosttraversalscorrectly.
SelectionofAppropriateTraversal	30	Correctly selects theappropriatetraversalfor every given problemscenario.	Selectsthecorrecttraversalformostscenarios.	Selectsappropriate traversalforonlyafewscenarios.	Frequentlyselectsincorrecttraversal methods.
Completion&TimeEfficiency	20	Completes all challengeswithin the allotted timewhilemaintainingaccuracy.	Completesmostchallengeswithinthe time limit.	Completesonlypartof the challenge orexceeds the timelimitslightly.	Unabletocompletethechallengewithin the allottedtime.
EvidenceSubmission	10	Uploadscompletescreenshotsshowingsuccessfulcompletion,score,andstudentdetails.	Uploadsscreenshotswithminormissinginformation.	Uploadsincomplete orunclearscreenshots.	Novalidevidencesubmitted.

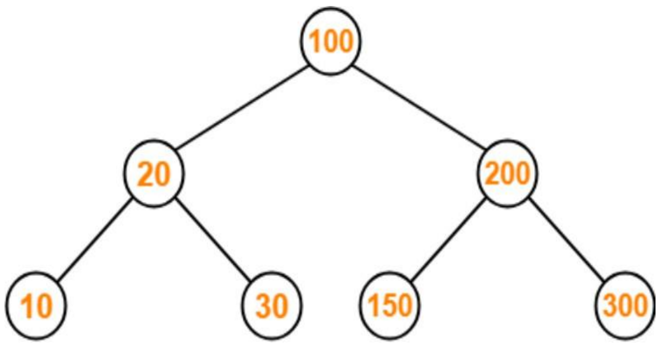
FacultyIn-charge	**CourseCoordinator**	**ProgramDirector**
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

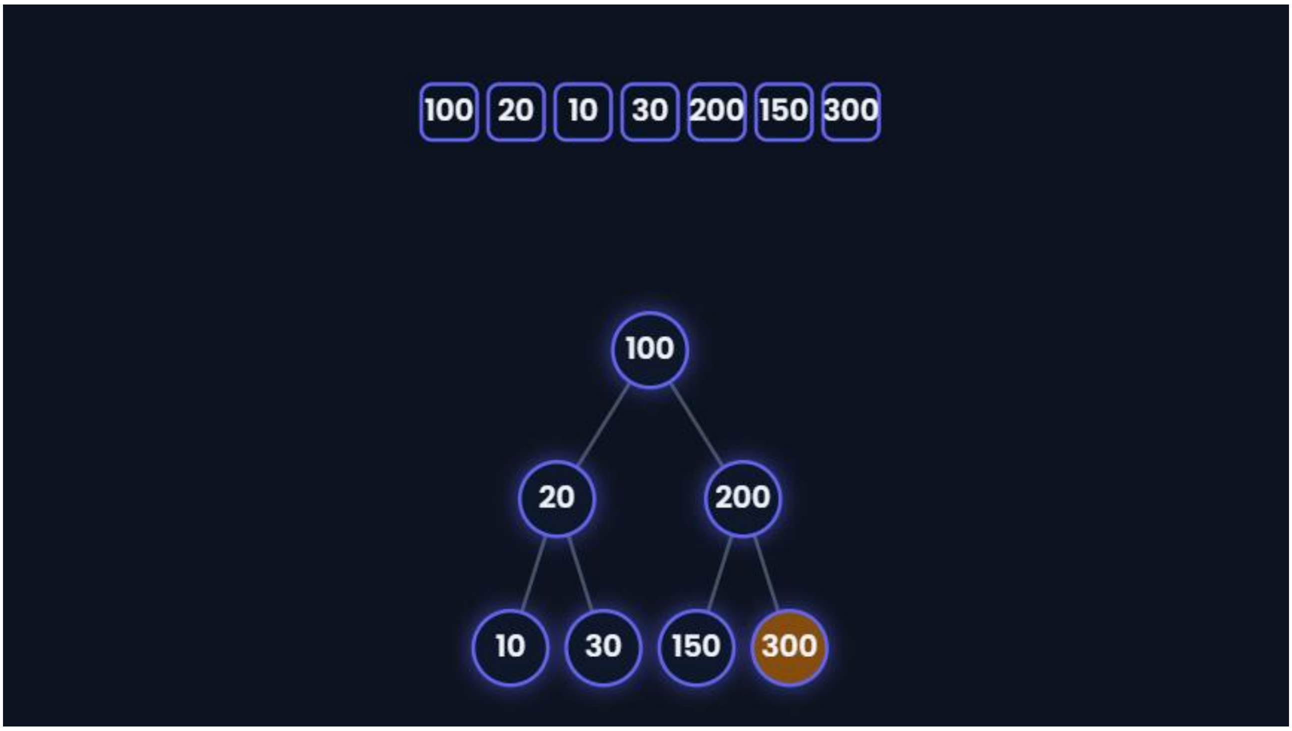
CSA0305 – Data Structures (Slot A) CO3-

AT2-TreeTraversalChallenges

QuestionNo:1

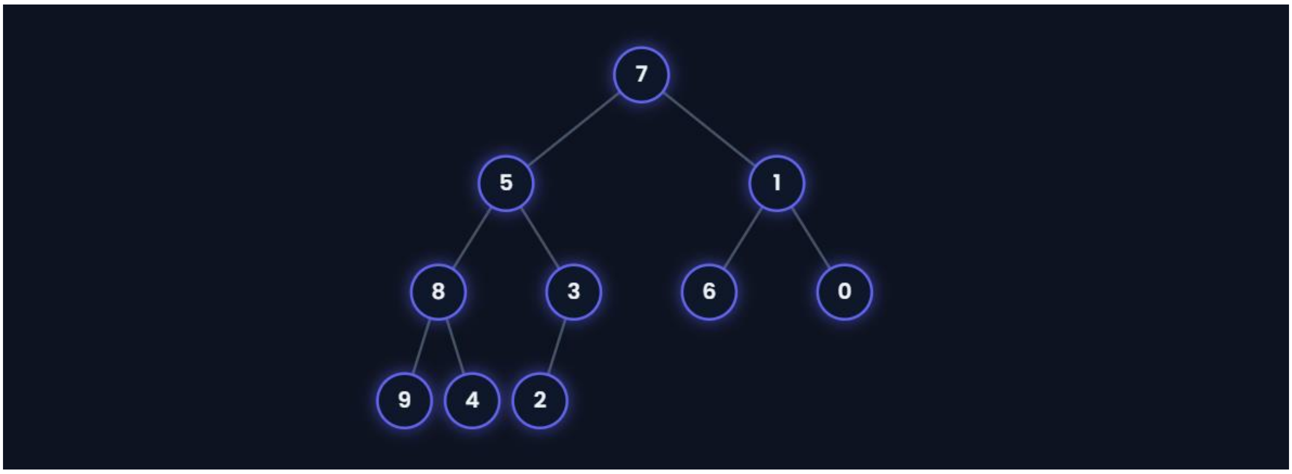
Writethetraversalsequencesforfollowingtrees





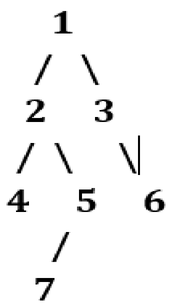
QuestionNo:2

Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a binary search tree that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?



QuestionNo:3

Write the Inorder, Preorder & Postorder traversal sequences for following trees



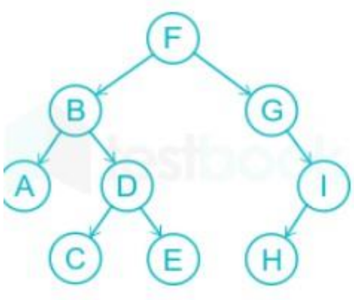
Pre order result [1,2,4,5,7,3,6]

Post order result [4,7,5,2,6,3,1]

In order result [4,2,7,5,1,3,6

QuestionNo:4

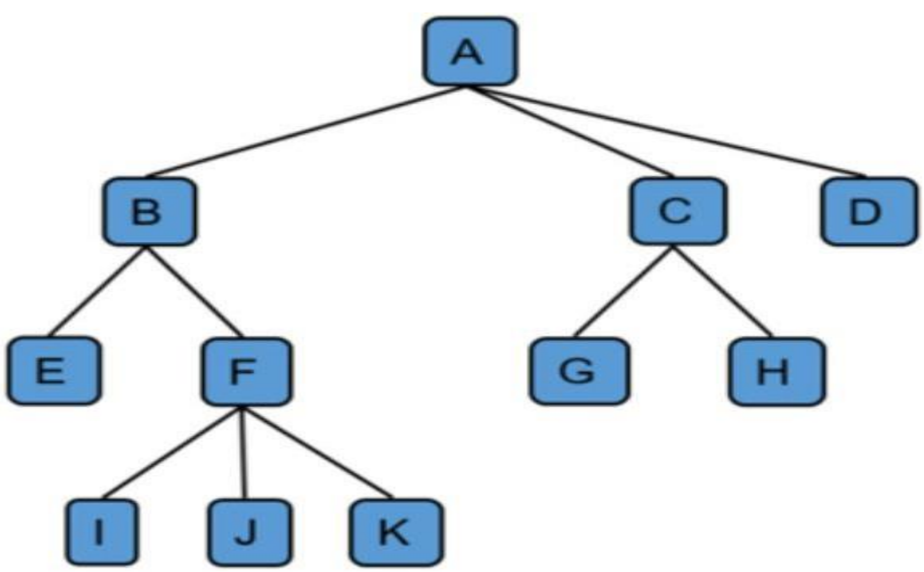
What is the sequence o f nodes when applying in-order traversal on the binary tree given below?



1. A, B, C, D, E, F, I, H, G
2. A, C, D, E, B, F, G, H, I
3. A, B, C, D, E, F, G, H, I
4. I, H, G, F, E, D, C, B, A

Answer [A,B,C,D,E,F,G,H,I]

QuestionNo:5



Preorder: A B E F I J K C G H D
Postorder: E I J K F B G H C D A